

acceleration remains constant.

5

Ans: **D**

collision:

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2 \quad \text{where } m_1 = m_2$$

$$u_1 + u_2 = v_1 + v_2$$

$$(+50) + (-30) = v_1 + v_2$$

$$v_1 + v_2 = 20$$

elastic

$$u_1 - u_2 = v_2 - v_1$$

$$(+50) - (-30) = v_2 - v_1$$

$$v_2 - v_1 = 80$$

	$v_1 + v_2$	$v_2 - v_1$
A	$0 + 20 = 20$	$20 - 0 = 20$
B	$10 + 10 = 20$	$10 - 10 = 0$
C	$(-20) + 0 = -20$	$0 - (-20) = 20$
D	$(-30) + (50) = 20$	$50 - (-30) = 80$

6

Ans: **B**